

STAFFORD, VA, USA

WMO: 720317

Lat: 38.398N

Lon: 77.456W

Elev: 65

StdP: 100.55

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03735

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-7.7	-19.5	0.7	-6.3	-16.7	0.9	-4.1	11.2	1.3	10.1	1.1	1.5	320	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.8	24.7	32.9	24.1	32.1	23.5	27.0	31.0	25.9	30.0	25.2	29.3	2.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	28.7	25.1	20.3	27.5	24.1	19.1	26.6	85.4	31.3	80.6	29.6	77.0	29.1	33.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.4	5.9	-15.6	36.4	3.2	1.4	-17.9	37.4	-19.8	38.2	-21.6	39.0	-23.9	40.0		
(5)			-16.0	28.4	2.3	1.3	-17.7	29.3	-19.0	30.1	-20.3	30.8	-22.0	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	2.1	3.5	7.7	13.7	18.4	22.8	25.2	24.3	20.7	14.2	8.1	4.3
	DBStd	9.31	5.82	5.71	5.56	5.00	4.40	3.20	2.58	2.85	3.60	4.75	4.61	4.92
	HDD10.0	876	255	194	114	18	1	0	0	0	0	13	91	190
	HDD18.3	2373	504	414	333	156	57	3	0	1	144	307	436	
	CDD10.0	2263	9	13	42	129	260	385	471	443	320	145	33	12
	CDD18.3	719	0	1	2	17	57	138	213	186	87	18	1	0
	CDH23.3	7001	0	3	34	217	582	1367	2124	1752	764	152	5	0
	CDH26.7	2769	0	0	3	58	178	527	952	744	271	36	0	0
(14) Wind	WSAvg	2.0	2.2	2.4	2.8	2.6	2.0	1.7	1.4	1.4	1.6	1.8	2.0	1.9
(15) Precipitation	PrecAvg	1047	81	74	97	77	99	85	104	89	97	89	84	85
	PrecMax	1516	192	162	243	177	219	274	297	254	278	202	184	180
	PrecMin	706	9	7	25	10	16	18	20	13	13	4	5	18
	PrecStd	206	38	41	47	36	49	50	66	58	63	50	48	47
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.8	27.1	31.4	32.6	35.6	36.5	36.2	33.9	30.3	23.9	21.3
		MCWB	14.7	15.9	16.8	20.5	22.8	25.1	25.5	24.5	24.1	22.9	16.9	17.0
	2%	DB	17.7	19.5	23.8	27.9	30.9	32.9	34.9	34.1	32.3	27.2	21.5	18.4
		MCWB	13.5	14.0	15.7	17.7	22.6	24.0	25.0	24.6	22.7	20.0	16.0	16.5
	5%	DB	14.9	16.0	20.8	26.0	28.7	31.8	32.9	32.4	30.1	24.8	19.2	15.6
		MCWB	11.2	10.8	14.2	16.9	21.1	23.4	24.9	23.5	22.6	19.6	14.4	12.6
	10%	DB	11.4	12.8	17.6	23.2	27.0	30.1	32.0	31.1	27.5	22.6	16.6	12.6
		MCWB	8.1	8.6	11.8	15.9	20.3	23.1	24.2	23.5	20.7	18.6	12.4	9.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	17.9	19.1	22.9	25.8	27.4	28.9	27.9	26.3	24.6	19.2	18.7
		MCDB	18.3	20.6	24.2	27.8	29.1	31.7	33.1	31.2	29.4	26.6	22.1	20.0
	2%	WB	14.1	15.1	16.9	20.6	24.1	26.0	27.2	26.4	25.1	22.8	17.3	16.7
		MCDB	17.7	18.2	21.7	24.7	27.6	30.3	31.5	29.8	28.8	25.0	19.8	18.0
	5%	WB	11.4	12.0	15.1	18.6	22.9	25.0	26.1	25.5	24.1	21.2	15.3	13.4
		MCDB	14.5	14.2	19.1	23.7	26.5	29.4	30.5	29.5	27.7	23.8	18.1	14.9
	10%	WB	8.4	9.1	12.9	17.0	21.8	24.1	25.2	24.8	22.9	19.3	13.1	10.2
		MCDB	10.6	12.7	17.3	22.2	25.4	28.4	29.4	28.9	25.7	21.8	15.8	11.9
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	12.1	13.6	12.5	12.2	11.7	11.8	11.8	12.2	12.2	10.5
		MCDBR	14.9	15.3	16.0	16.3	14.5	14.2	13.4	13.6	14.6	14.1	15.4	13.4
	5% WB	MCWBR	11.6	10.6	9.6	7.9	7.0	6.8	5.6	5.5	6.5	8.0	9.9	10.5
		MCWBR	13.3	13.4	13.7	13.7	12.1	12.2	11.9	11.4	11.5	11.4	11.9	12.1
(40) Clear-Sky Solar Irradiance	taub		0.303	0.317	0.352	0.395	0.433	0.488	0.507	0.507	0.417	0.356	0.321	0.307
		taud	2.507	2.447	2.404	2.304	2.223	2.109	2.079	2.059	2.293	2.463	2.520	2.532
	Ebn at Noon		882	911	908	884	852	802	783	771	828	855	852	852
		Edh at Noon	78	94	108	127	141	158	161	161	120	92	77	71
(44) All-Sky Solar Radiation	RadAvg		2.16	2.97	3.96	4.93	5.58	6.13	5.96	5.34	4.37	3.34	2.45	1.91
	RadStd		0.19	0.33	0.28	0.36	0.58	0.46	0.31	0.26	0.51	0.38	0.20	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	+135	+82

Nomenclature: See separate page